Energy Based Models Don't Generalize like Diffusion Models

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1 Introduction

A modern deep network often has far more parameters than training examples and yet generalises well. In principle, such a model could simply memorise its training set; in practice, stochastic gradient descent (SGD) frequently discovers solutions that extrapolate meaningfully beyond the data. Why? This gap between capacity and behavior has motivated a long line of work on implicit regularization, ranging from kernel limits and minimum-norm biases to data-dependent effects such as low-dimensional structure and spectral biases [7, 4, 1, 12].

A complementary perspective treats generalization as a dynamical phenomenon: models learn different patterns in data at different rates. For example, in many regression problems such as ridge regularization, training first moves along directions that fit broad patterns, and later along directions that fit sample-specific noise. The reverse order can also occur: models can overfit before they generalize, and only later "grok" the underlying rule [8]. These phenomena suggest that learning can be characterized by two timescales:τ_{gen}, the minimum training time required to achieve generalization (eg. high-quality image generation), and τ_{mem}, the minimum training time required to memorize the given dataset (i.e. near-zero training loss). These two timescales are factors of parameter count, model architecture, data size, and data complexity (i.e. number of bits needed to describe data).

This perspective clarifies why diffusion and score-based generative models often resist memorization even when highly overparameterized. Bonnaire et al. identify an "implicit dynamical regularization" mechanism in diffusion training: the memorization timescale τ_{mem} grows linearly with dataset size n while τ_{gen} is constant. This creates a $O(n)$ window where training improves sample quality while avoiding memorization [2].

In this paper, we ask whether this mechanism persists for energy-based models (EBMs). Both diffusion models and EBMs can be trained with score matching objectives [6, 13, 10]. Similarities like this motivate the question: do EBMs and diffusion models share the same dynamical regularization preventing memorization?

Surprisingly, we find that EBMs exhibit substantially different behavior. Although both diffusion models and EBMs learn s_θ through the same score-matching loss, EBMs have the additional constraint that $s_\theta = -\nabla_x E_\theta$. That is, s_θ needs to be a conservative vector field. This has a substantial impact on learning dynamics: it removes the $O(n)$ timescale τ_{mem} we observe in diffusion models.

We validate this phenomenon both theoretically and empirically. Using a random feature model, we prove that with diffusion, a feature covariance matrix exhibits two bulks in its spectrum, but one of the bulks is neglected when we switch to EBMs. Next, we verify our theory using numerical simulations, and correctly predict both the desired spectrum and training rates. Finally, we train a deep EBM on real image datasets, and demonstrate find that the gap $\tau_{mem} - \tau_{gen}$ indeed vanishes compared to diffusion models [2]. Together, these results suggest that the conservative constraint in EBMs remove a key source of implicit dynamical regularization present in diffusion training, with concrete consequences for how generalization windows scale with data.

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2 Related Work

Diffusion and score-based generative models. Diffusion probabilistic models learn to reverse a progressive noising process, starting from early physics-inspired formulations [9] and modern DDPM training objectives [5]. Score-based generative modeling frames sampling as integrating reverse-time SDEs driven by a learned time-dependent score [11]. These lines connect to EBMs because the score is a conservative field and can be written as the gradient of a (time-dependent) energy.

Overfitting, early stopping, and learning timescales. Early stopping acts as an implicit regularizer in iterative algorithms, classically understood via a spectral bias-variance tradeoff in kernel regression and inverse problems [14]. Modern overparameterized regimes exhibit additional phenomena such as double descent and benign interpolation, which sharpen the distinction between fitting and generalizing [1, 4]. In contrast, grokking highlights delayed generalization, where optimization first achieves low training error and only later generalizes [8]. These works motivate describing training by multiple timescales rather than a single convergence rate.

Spectral theories of overfitting in EBMs and diffusion. Our work is inspired by [2], which attributes the generalization–memorization transition in diffusion training to an implicit dynamical regularization mechanism and explains it with a random-features theory where eigenvalues of a feature covariance matrix govern relaxation rates. On the EBM side, [3] develop a spectral framework for overfitting in (pairwise) EBMs, dissecting learning trajectories across eigenmodes of an empirical covariance and connecting early stopping to finite-sample spectral corrections. Our contribution differs in focus and mechanism: we study conservative score parameterizations and show that conservativity induces a Hadamard-lifted kernel, which can collapse the spectral structure that drives slow memorization in diffusion score models.

3 Preliminaries

In this section, we introduce denoising score matching (DSM), the score-based training principle we will assume for our energy-based models throughout the paper. We then review Bonnaire et al.'s framework for analyzing random-feature score models, emphasizing the specific parts from their paper we will adapt. Finally, we record the random-matrix identities and concentration tools that we will use in our analysis. The key point is that the conservative kernel of our RF-EBM will be a Hadamard lift of Bonnaire et al.'s feature covariance matrix, and these Hadamard products can be rewritten as Gram matrices in a lifted dimension whose spectra we can control in the proportional regime.

3.1 Denoising score matching (DSM)

Let $x \in \mathbb{R}^d$ be drawn from a target data distribution P_x and let $\xi \sim \mathcal{N}(0, I_d)$ be independent noise. Fix a noise level $\tau > 0$ and define the corrupted sample

$$y \;=\; x + \sqrt{\tau}\,\xi.$$

Let P_τ denote the law of y, i.e. $P_\tau = P_x * \mathcal{N}(0, \tau I_d)$, and let $s_\tau : \mathbb{R}^d \to \mathbb{R}^d$ denote its score,

$$s_\tau(y) \;:=\; \nabla_y \log p_\tau(y),$$

where p_τ is the density of P_τ (when it exists).

Definition 3.1 (Denoising score matching)**.** Given a parametric score model $\psi_\theta : \mathbb{R}^d \to \mathbb{R}^d$, DSM trains ψ_θ by minimizing

$$L_{\mathrm{DSM}}(\theta) := \frac{1}{d}\,\mathbb{E}_{x,\xi}\Big\|\psi_\theta(x + \sqrt{\tau}\,\xi) + \frac{1}{\sqrt{\tau}}\xi\Big\|_2^2. \tag{1}$$

3.2 Unconstrained random-feature score models

We now summarize the random-feature score-learning framework used by Bonnaire et al. The point is to isolate the methods we will adapt (quadratic DSM $\Rightarrow$ linear gradient flow $\Rightarrow$ spectral analysis of fitting rates), and to make clear where their analysis relies on the fact that the score model is unconstrained (unlike an EBM score, which is conservative).

3.2.1 Model class and the role of random features

Fix a frozen random feature matrix $W \in \mathbb{R}^{p\times d}$ with i.i.d. rows $w_i^\top \sim \mathcal{N}(0, I_d)$ and a scalar nonlinearity $\sigma : \mathbb{R} \to \mathbb{R}$. The feature vector $\phi(x) := \sigma(Wx/\sqrt{d}) \in \mathbb{R}^p$ should be thought of as a one-hidden-layer random network with width p and fixed first-layer weights. The learnable part of the model is an output map that converts these p scalar features into a d-dimensional score field.

Bonnaire et al. consider the unconstrained score class

$$s_A(x) \;:=\; \frac{1}{\sqrt{p}}\, A\,\phi(x), \qquad A \in \mathbb{R}^{d\times p} \text{ trainable.} \tag{2}$$

Bonnaire et al. write DSM at a fixed diffusion time t using an Ornstein–Uhlenbeck forward process, so their noise level appears as $\Delta_t = 1 - e^{-2t}$. In this paper we use the equivalent additive-noise parameterization $y = x + \sqrt{\tau}\xi$.

3.2.2 Quadratic DSM loss and closed-form gradient flow

Let $\{x^\nu\}_{\nu=1}^n$ be training data and $y^\nu = x^\nu + \sqrt{\tau}\xi^\nu$ their noised versions. Define

$$\phi^\nu := \phi(y^\nu) = \sigma(Wy^\nu/\sqrt{d}) \in \mathbb{R}^p,$$

and define the (noise-averaged) feature covariance and feature–noise cross-covariance

$$U \;:=\; \frac{1}{n}\sum_{\nu=1}^{n} \mathbb{E}_\xi\big[\phi^\nu(\phi^\nu)^\top\big] \in \mathbb{R}^{p\times p}, \qquad V \;:=\; \frac{1}{n}\sum_{\nu=1}^{n} \mathbb{E}_\xi\big[\phi^\nu(\xi^\nu)^\top\big] \in \mathbb{R}^{p\times d}. \tag{3}$$

A direct expansion of the DSM square yields an exact quadratic decomposition in A:

$$L_{\text{train}}(A) = \frac{1}{\tau} + \frac{1}{pd}\mathrm{Tr}(AUA^\top) + \frac{2}{d\sqrt{p\tau}}\mathrm{Tr}(AV) \tag{4}$$

(up to the same harmless deterministic rescalings between the OU and additive-noise normalizations).

The quadratic form (4) implies that full-batch gradient descent admits a closed form. If we take the rescaled gradient flow

$$\dot{A}(t) = -p\nabla_A L_{\text{train}}(A(t)),$$

then differentiating (4) gives

$$\dot{A}(t) = -\frac{2}{d}A(t)U - \frac{2\sqrt{p}}{d\sqrt{\tau}}\,V^\top. \tag{5}$$

Assuming U is invertible, the unique fixed point is $A^\star = -(\sqrt{p}/\sqrt{\tau})\,V^\top U^{-1}$, and the solution is

$$A(t) = A^\star + \big(A(0) - A^\star\big)\exp\Big(-\frac{2}{d}Ut\Big). \tag{6}$$

If $Uv_k = \lambda_k v_k$, then the component of $A(t) - A^\star$ along v_k decays like $\exp(-(2/d)\lambda_k t)$. This tells us that the relaxation rates are controlled by the eigenvalues of U (up to the deterministic prefactor $2/d$ coming from our normalization).

3.2.3 Gaussian equivalence and the MP band in Bonnaire et al.

Bonnaire et al. invoke a Gaussian equivalence principle to obtain an explicit surrogate for U.

Lemma 3.2 (Gaussian-equivalent surrogate for the feature covariance U)**.** *Assume: (i) σ admits a Hermite expansion with finite moments, and (ii) the data distribution is sub-Gaussian with covariance Σ_x of bounded operator norm and $\mathrm{Tr}(\Sigma_x)/d \to \sigma_x^2 \in (0,\infty)$, and (iii) $d, p, n \to \infty$ with $\psi_p := p/d$ and $\psi_n := n/d$ finite. Then at any fixed noise level, U is asymptotically equivalent (in empirical spectral distribution, equivalently in Stieltjes transforms at fixed $z \in \mathcal{C}^+$) to a Gaussian surrogate of the form*

$$U_G \;=\; s^2 I_p \;+\; \frac{v^2}{n}\,\Omega\Omega^\top \;+\; b^2\,\frac{WW^\top}{d}, \tag{7}$$

where $\Omega \in \mathbb{R}^{p\times n}$ has i.i.d. $\mathcal{N}(0,1)$ entries independent of W, and $s^2, v^2, b^2 \geq 0$ are deterministic constants depending on (σ, τ, Σ_x). (If $\mathbb{E}[\sigma(\cdot)] \neq 0$, an additional rank-one term appears. It only produces an outlier, as we see in Figure 1)

The key point is that the finite-n term $(1/n)\Omega\Omega^\top$ has a nontrivial limiting spectrum when $p/n \to \gamma \in (0, \infty)$. For convenience, we record the standard Marchenko–Pastur theorem in the form we use to interpret this term.

Theorem 3.3 (Marchenko–Pastur law, sample covariance form)**.** *Let $Z \in \mathbb{R}^{p\times n}$ have i.i.d. entries with mean 0, variance 1, and finite fourth moment, and assume $p/n \to \gamma \in (0, \infty)$. Then the empirical spectral distribution of $(1/n)ZZ^\top$ converges weakly almost surely to $\mathrm{MP}(\gamma)$, supported on $[(1-\sqrt{\gamma})^2, (1+\sqrt{\gamma})^2]$ (with an atom at 0 of mass $1 - 1/\gamma$ when $\gamma > 1$).*

Combining Lemma 3.2 with Theorem 3.3 yields the qualitative two-band picture in Bonnaire et al.: one part of the spectrum comes from the finite-n MP band of $(1/n)\Omega\Omega^\top$, while another part comes from the population-dependent term $WW^\top/d$. Since the gradient flow (6) relaxes along eigenmodes of U, this spectral separation induces two distinct fitting timescales in their analysis. Whether this MP-driven mechanism persists after imposing conservativity is the main question addressed in Section 4.

The unconstrained model (2) has a key limitation that prevent us from directly transferring Bonnaire et al.'s spectrl analysis to random-feature EBMs. An EBM score is a gradient field: $\psi(x) = -\nabla_x E(x)$ for some scalar energy E. A generic random-feature score of the form $A\,\sigma(Wx)$ is not conservative, so the kernel U controlling learning in diffusion models is not the kernel that arises for EBMs trained by DSM.

This limitation motivate our main technical task in Section 4: once conservativity is enforced, we must redo the spectral analysis for the new kernel that replaces U, and then reinterpret the resulting eigenstructure in terms of training dynamics and early stopping.

3.3 Other technical tools

Our analysis of conservative random-feature EBMs reduces to controlling the spectrum of Hadamard-lifted matrices such as $A \odot (\Omega\Omega^\top/n)$ and $A^{\odot 2}$ (and their bounded-covariance analogues). Hence here we record the algebraic lift identities that rewrite these Hadamard products as Gram matrices in a higher dimension, and the concentration results that control their spectra in the regimes we study.

3.3.1 Schur–Khatri–Rao lift: Hadamard products of Grams are Grams

The following identity is the algebraic reason Hadamard products simplify in our setting.

Lemma 3.4 (Schur product of Gram matrices is a Gram matrix)**.** *Let $U \in \mathbb{R}^{p\times D_1}$ and $V \in \mathbb{R}^{p\times D_2}$ have rows $u_i^\top$ and $v_i^\top$. Define the Gram matrices*

$$G_U := \frac{1}{D_1} UU^\top, \quad G_V := \frac{1}{D_2} VV^\top.$$

Let $Z \in \mathbb{R}^{p\times(D_1 D_2)}$ be the matrix whose ith row is $z_i^\top := u_i^\top \otimes v_i^\top$. Then

$$G_U \odot G_V \;=\; \frac{1}{D_1 D_2}\, ZZ^\top. \tag{8}$$

Proof. Entrywise, we have

$$(G_U \odot G_V)_{ij} = \left(\tfrac{1}{D_1} u_i^\top u_j\right)\left(\tfrac{1}{D_2} v_i^\top v_j\right) = \tfrac{1}{D_1 D_2}(u_i \otimes v_i)^\top (u_j \otimes v_j) = \tfrac{1}{D_1 D_2}(ZZ^\top)_{ij},$$

so $G_U \odot G_V = (1/(D_1 D_2))ZZ^\top$, as claimed. □

In Section 4, the conservative kernel contains terms like $(WW^\top/d) \odot (\Omega\Omega^\top/n)$ and $(WW^\top/d)^{\odot 2}$. Lemma 3.4 rewrites these as sample covariances in lifted dimensions dn and d^2, respectively. This lift is the mechanism behind the spectral collapse we prove in the proportional regime.

3.3.2 Concentration of lifted Gram matrices when $p/D \to 0$

Once we rewrite a Hadamard-lifted term as $(1/D)ZZ^\top$, the relevant regime depends on p/D. In the proportional regime, the lifted dimension is much larger than p (e.g. $D = dn$ or $D = d^2$), and the Gram matrix concentrates near the identity.

Lemma 3.5 (Operator-norm concentration when $p/D \to 0$)**.** *Suppose we have a matrix $Z \in \mathbb{R}^{p\times D}$ with independent rows $z_i^\top$ satisfying:*

- *Isotropy. $\mathbb{E}[z_i] = 0$ and $\mathbb{E}[z_i z_i^\top] = I_D$;*
- *Sup-exponential bounded norm. $\|z_i\|_{\psi_1} \le K$ uniformly in i.*

Then there exist constants $c, C > 0$ (depending only on K) such that for all $t \ge 0$, with probability at least $1 - 2e^{-ct^2}$,

$$\Big\|\frac{1}{D}ZZ^\top - I_p\Big\|_{\mathrm{op}} \;\le\; C\Big(\sqrt{\frac{p}{D}} + \frac{t}{\sqrt{D}}\Big) + C^2\Big(\frac{p}{D} + \frac{t^2}{D}\Big). \tag{9}$$

In particular, if $p/D \to 0$ then $\|(1/D)ZZ^\top - I_p\|_{\mathrm{op}} \to 0$ in probability.

In Section 4, after applying Lemma 3.4, we will be in exactly the regime $p/(dn) \to 0$ and $p/d^2 \to 0$ (since $p \asymp d$ and $n \asymp d$). Lemma 3.5 then explains why the MP-scale fluctuations present in $(1/n)\Omega\Omega^\top$ collapse after the conservative Hadamard lift.

4 Conservative random-feature EBMs in the proportional regime

Our goal in this section is to see whether Bonnaire et al.'s two-timescale mechanism survives once we impose the structural constraint that is intrinsic to energy-based models: the learned score must be a gradient field, $\psi_\theta(x) = -\nabla_x E_\theta(x)$.We state the results and the proof ideas here, and defer complete proofs to the appendix.

4.1 Setup

Fix a differentiable scalar potential $\Phi : \mathbb{R} \to \mathbb{R}$ and write $\sigma := \Phi'$. This is our scalar non-linear activation function. We consider the random-feature energy family, linear in $\theta \in \mathbb{R}^p$,

$$E_\theta(x) := -\frac{\sqrt{d}}{\sqrt{p}} \sum_{i=1}^{p} \theta_i\, \Phi\Big(\frac{w_i^\top x}{\sqrt{d}}\Big), \tag{10}$$

with score field

$$\psi_\theta(x) := -\nabla_x E_\theta(x) = \frac{1}{\sqrt{p}}\, W^\top\Big(\theta \odot \sigma(Wx/\sqrt{d})\Big) \in \mathbb{R}^d. \tag{11}$$

The key point relative to Bonnaire et al. is that the output map is not free: the architecture ties it to W in the way needed to ensure ψ_θ is conservative. We train θ by the DSM objective at noise level τ,

$$L_{\mathrm{train}}(\theta) := \frac{1}{dn}\sum_{\nu=1}^{n} \mathbb{E}_{\xi^\nu}\Big\|\psi_\theta(y^\nu) + \frac{1}{\sqrt{\tau}}\xi^\nu\Big\|_2^2. \tag{12}$$

4.2 Exact quadratic reduction of DSM and the conservative kernel

Define the feature vector at sample ν by

$$\phi^\nu := \sigma(Wy^\nu/\sqrt{d}) \in \mathbb{R}^p, \tag{13}$$

and the empirical (noise-averaged) feature covariance

$$U_{\mathrm{train}} := \frac{1}{n}\sum_{\nu=1}^{n} \mathbb{E}_{\xi^\nu}\big[\phi^\nu(\phi^\nu)^\top\big] \in \mathbb{R}^{p\times p}. \tag{14}$$

Proposition 4.1 (Quadratic reduction and the conservative training kernel)**.** *The DSM objective* (12) *admits the exact decomposition*

$$L_{\text{train}}(\theta) = \frac{1}{\tau} + \frac{1}{p}\,\theta^\top H_{\text{train}}\theta + \frac{2}{\sqrt{p}}\,\theta^\top b_{\text{train}}, \tag{15}$$

where the conservative kernel is the Hadamard lift

$$H_{\text{train}} := A \odot U_{\text{train}}, \qquad A = \frac{1}{d}WW^\top, \tag{16}$$

and the linear term $b_{\text{train}} \in \mathbb{R}^p$ *is*

$$(b_{\text{train}})_i := \frac{1}{nd\sqrt{\tau}} \sum_{\nu=1}^{n} \mathbb{E}_{\xi^\nu}\Big[\phi_i^\nu\, w_i^\top \xi^\nu\Big]. \tag{17}$$

Proof sketch. Expand the square in (12). The constant term satisfies $\mathbb{E}\|\xi\|^2/d = 1$, hence contributes $1/\tau$. For the quadratic term, use (11) to write $\psi_\theta(y) = \frac{1}{\sqrt{p}}W^\top \text{diag}(\phi)\theta$, hence

$$\|\psi_\theta(y)\|^2 = \frac{1}{p}\theta^\top \text{diag}(\phi)\, WW^\top\, \text{diag}(\phi)\theta.$$

The entrywise identity $\text{diag}(\phi)\, M\, \text{diag}(\phi) = M \odot (\phi\phi^\top)$ then produces $A \odot (\phi\phi^\top)$, and averaging over ν and ξ^ν yields $A \odot U_{\text{train}}$. The linear term comes from the cross-term $\xi^\top \psi_\theta(y)$ and gives (17). Full details appear in Appendix. □

In the unconstrained RF score model of Bonnaire et al., the effective dynamics kernel is U itself. Here, the conservative parameter tying introduces an additional Gram factor A and forces the kernel to be the Schur product $H_{\text{train}} = A \odot U_{\text{train}}$. The rest of the section is an analysis of the spectrum of this lifted kernel in the proportional regime.

4.3 Spectral collapse of the conservative kernel in the proportional regime

We now state the main conclusion of this section: in the proportional regime with $p \asymp d$ and $n \asymp d$, the conservative kernel $H_{\text{train}} = A \odot U_{\text{train}}$ does not inherit the finite-n Marchenko–Pastur bulk that drives the slow phase in the unconstrained RF score model.

Theorem 4.2 (Operator-norm limit of the conservative kernel in the proportional regime)**.** *Assume our data is generated by an isotropic Gaussian model* $x \sim \mathcal{N}(0, \sigma_x^2 I_d)$ *and that we are in the proportional regime. Assume the Gaussian-equivalent surrogate for* U_{train} *from Section 3 applies at the fixed noise level* τ*, so that there exist deterministic constants* $s^2, v^2, b^2 \geq 0$ *and an auxiliary matrix* $\Omega \in \mathbb{R}^{p\times n}$ *with i.i.d.* $\mathcal{N}(0,1)$ *entries independent of* W *such that*

$$U_{\text{train}} \approx s^2 I_p \;+\; v^2\,\frac{1}{n}\Omega\Omega^\top \;+\; b^2 A, \qquad A = \frac{1}{d}WW^\top, \tag{18}$$

up to negligible errors in operator norm.[1] *Then*

$$\Big\| H_{\text{train}} - \Big((s^2+v^2+b^2)I_p + \frac{b^2}{d}\mathbf{1}\mathbf{1}^\top\Big)\Big\|_{\text{op}} \xrightarrow{\mathbb{P}} 0 \tag{19}$$

as $d \to \infty$.

Proof Sketch. Using $H_{\text{train}} = A \odot U_{\text{train}}$ and (18), we have

$$H_{\text{train}} \approx s^2(A \odot I_p) \;+\; v^2\Big(A \odot \frac{1}{n}\Omega\Omega^\top\Big) \;+\; b^2(A \odot A).$$

Here $A \odot I_p = \text{diag}(A)$ and $A \odot A = A^{\odot 2}$. First, by χ^2 concentration and a union bound over $i \leq p$,

$$\|\text{diag}(A) - I_p\|_{\text{op}} \xrightarrow{\mathbb{P}} 0,$$

[1] If $\mathbb{E}[\sigma(\Gamma g)] \neq 0$, an additional rank-one term appears in (18); it only affects a single outlier direction and does not change the bulk conclusions below.

so $s^2(A \odot I_p) = s^2 I_p + o_{\mathbb{P}}(1)$. Next, by the Schur–Gram identity (Lemma 3.4),

$$A \odot \frac{1}{n}\Omega\Omega^\top \;=\; \frac{1}{dn} Z Z^\top, \qquad z_i = w_i \otimes \omega_i \in \mathbb{R}^{dn}.$$

The rows z_i are i.i.d. across i, isotropic in $\mathbb{R}^{dn}$, and have sub-exponential coordinates (products of Gaussians), so Lemma 3.5 applies with $D = dn$; since $p/(dn) \to 0$ in the proportional regime,

$$\Big\| A \odot \frac{1}{n}\Omega\Omega^\top - I_p \Big\|_{\mathrm{op}} \xrightarrow{\mathbb{P}} 0,$$

and therefore $v^2\big(A \odot (\Omega\Omega^\top/n)\big) = v^2 I_p + o_{\mathbb{P}}(1)$. Finally, letting $q_i := w_i \otimes w_i \in \mathbb{R}^{d^2}$,

$$(A^{\odot 2})_{ij} = \Big(\frac{w_i^\top w_j}{d}\Big)^2 = \frac{1}{d^2}\, q_i^\top q_j, \qquad \text{so } A^{\odot 2} = \frac{1}{d^2} Q Q^\top,$$

where $Q \in \mathbb{R}^{p \times d^2}$ have rows $q_i^\top$. Write $q_i = \mu + \tilde{q}_i$ with $\mathbb{E}[\tilde{q}_i] = 0$ so that

$$\frac{1}{d^2} Q Q^\top = \frac{1}{d^2}\tilde{Q}\tilde{Q}^\top + \frac{\|\mu\|^2}{d^2}\mathbf{1}\mathbf{1}^\top + \frac{1}{d^2}\Big(\tilde{Q}\mu\,\mathbf{1}^\top + \mathbf{1}(\tilde{Q}\mu)^\top\Big).$$

The centered rows $\tilde{q}_i$ have sub-exponential tails and after rescaling by their second moment they satisfy the hypotheses of Lemma 3.5 in $D = d^2$ with $p/D \to 0$, so $(1/d^2)\tilde{Q}\tilde{Q}^\top$ concentrates around $(1 + 1/d)I_p$. Then, because $\|\mu\|^2 = \|I_d\|_F^2 = d$, the middle term equals $(1/d)\mathbf{1}\mathbf{1}^\top$. The cross terms are negligible in operator norm because $\|\tilde{Q}\mu\|_2 = O_{\mathbb{P}}(\sqrt{p}\,\|\mu\|)$ while $p/(d^2) \to 0$ in the proportionality regime, so their contribution is $o_{\mathbb{P}}(1)$. Hence

$$A^{\odot 2} = \Big(1 + \frac{1}{d}\Big) I_p + \frac{1}{d}\mathbf{1}\mathbf{1}^\top + o_{\mathbb{P}}(1) = I_p + \frac{1}{d}\mathbf{1}\mathbf{1}^\top + o_{\mathbb{P}}(1).$$

Multiplying by b^2 gives $b^2 A^{\odot 2} = b^2 I_p + \frac{b^2}{d}\mathbf{1}\mathbf{1}^\top + o_{\mathbb{P}}(1)$. Substituting these three approximations back into the decomposition for H_{train} yields

$$H_{\mathrm{train}} = (s^2 + v^2 + b^2) I_p + \frac{b^2}{d}\mathbf{1}\mathbf{1}^\top + o_{\mathbb{P}}(1)$$

in operator norm, which is equivalent to (19). □

Corollary 4.3 (Bulk eigenvalue collapse and absence of a vanishing spectral band)**.** *Under the assumptions of Theorem 4.2, with probability tending to* 1*,*

$$\lambda_k(H_{\mathrm{train}}) = (s^2 + v^2 + b^2) + o(1) \quad \textit{for all but at most one } k,$$
$$\lambda_{\max}(H_{\mathrm{train}}) = (s^2 + v^2 + b^2) + b^2\psi_p + o(1).$$

In particular, if $s^2 + v^2 + b^2 > 0$, then $\lambda_{\min}(H_{\mathrm{train}}) \geq (s^2 + v^2 + b^2)/2$ for all sufficiently large d with high probability, so the bulk spectrum is bounded away from 0 *and there is no additional family of parametrically small eigenvalues generated by finite-n fluctuations.*

Proof Sketch. Writing $c := s^2 + v^2 + b^2$ and $R := \frac{b^2}{d}\mathbf{1}\mathbf{1}^\top$, by Weyl's inequality and Theorem 4.2,

$$\max_{1 \leq k \leq p} \big|\lambda_k(H_{\mathrm{train}}) - \lambda_k(cI_p + R)\big| \leq \|H_{\mathrm{train}} - (cI_p + R)\|_{\mathrm{op}} = o_{\mathbb{P}}(1).$$

The matrix R has rank one with eigenvalues $\lambda_{\max}(R) = \frac{b^2}{d}\|\mathbf{1}\|_2^2 = \frac{b^2 p}{d} = b^2\psi_p + o(1)$ and $\lambda_k(R) = 0$ for $k = 2, \ldots, p$. Hence the eigenvalues of $cI_p + R$ are

$$\lambda_{\max}(cI_p + R) = c + b^2\psi_p + o(1), \qquad \lambda_k(cI_p + R) = c \;\text{ for } k = 2, \ldots, p,$$

as desired. For the final claim, assume $c > 0$. Since $c > 0$, with probability tending to 1, we have

$$\big|u^\top\big(H_{\mathrm{train}} - (cI_p + R)\big)u\big| \leq \|H_{\mathrm{train}} - (cI_p + R)\|_{\mathrm{op}} \leq c/2,$$

for any $u \in \mathbb{R}^p$ with $\|u\|_2 = 1$ by the definition of the operator norm and Theorem 4.2, so $u^\top H_{\mathrm{train}} u \geq u^\top(cI_p + R)u - c/2$. Since $R \succeq 0$, we have $u^\top(cI_p + R)u \geq c$ for all unit u, so

$$\lambda_{\min}(H_{\mathrm{train}}) = \min_{\|u\|_2 = 1} u^\top H_{\mathrm{train}} u \geq c - c/2 = c/2 > 0$$

almost surely by the Rayleigh quotient characterization, which is what we wanted to show. □

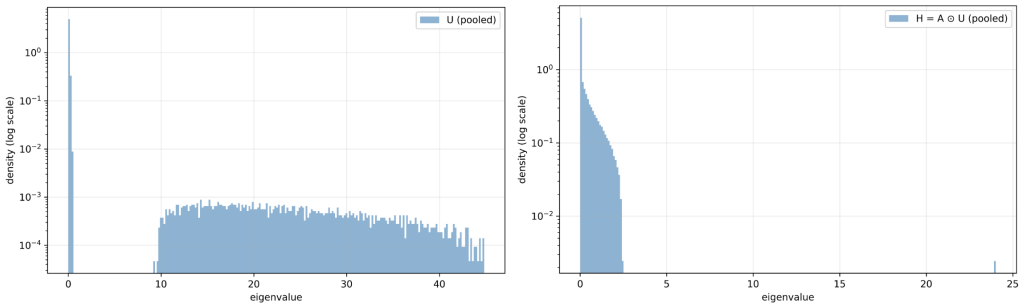

Figure 1: **(Left)** Diffusion kernel U: two-scale spectrum in the overparameterized regime (following [2]). The finite-n component yields an MP-like band near 0 while the population component produces a separated bulk. **(Right)** EBM kernel H: the conservative kernel $H = A \odot U$ concentrates to a narrow band, consistent with Corollary 4.3.

4.4 Training dynamics: what the collapsed spectrum implies for early stopping

We now interpret Corollary 4.3 at the level of training dynamics. Because our reduction (15) is quadratic, full-batch gradient descent is exactly a linear dynamical system. Concretely, if we run gradient descent with step size η, we get

$$\theta^{(k+1)} = \theta^{(k)} - \eta \nabla_\theta L_{\text{train}}(\theta^{(k)}).$$

Taking the small-step limit $\eta \to 0$ while rescaling time as $t := k\eta/p$ yields the gradient flow

$$\dot{\theta}(t) = -p\nabla_\theta L_{\text{train}}(\theta(t)) = -2H_{\text{train}}\theta(t) - 2\sqrt{p}\, b_{\text{train}}. \tag{20}$$

When H_{train} is invertible, the solution is

$$\theta(t) = \theta^\star + e^{-2H_{\text{train}}t}\big(\theta(0) - \theta^\star\big), \qquad \theta^\star = -\sqrt{p}\, H_{\text{train}}^{-1} b_{\text{train}}. \tag{21}$$

If we diagonalize $H_{\text{train}} v_k = \lambda_k v_k$ and write $b_{\text{train}} = \sum_k \beta_k v_k$, then along each eigenmode

$$\langle \theta(t) - \theta^\star, v_k \rangle = e^{-2\lambda_k t} \langle \theta(0) - \theta^\star, v_k \rangle, \qquad \langle \theta^\star, v_k \rangle = -\sqrt{p}\, \frac{\beta_k}{\lambda_k}.$$

Thus the only way to obtain a parametrically slow phase in this linearized dynamics is for H_{train} to possess a nontrivial band of eigenvalues near 0 (and for b_{train} to have non-negligible projection onto the corresponding eigenspace). Corollary 4.3 rules out exactly this scenario in proportional scaling: apart from a possible rank-one outlier direction, all eigenvalues concentrate near the constant $c = s^2 + v^2 + b^2$, and in particular $\lambda_{\min}(H_{\text{train}})$ is bounded away from 0 whenever $c > 0$. Consequently, in the rescaled time variable t, the linearized training dynamics relaxes on a single bulk time scale of order c^{-1} (with at most one additional direction that relaxes even faster, corresponding to the outlier eigenvalue).

Implication for early stopping. In the random feature diffusion model, the finite-n Wishart component produces a MP bulk at aspect ratio p/n. Particularly, the bottom edge of that bulk can be small, which creates a band of slow modes in U and hence a long transient window where the high-eigenvalue modes have already equilibrated while the low-edge modes remain effectively frozen. This spectral separation is what produces the early-stopping window $\tau_{mem} - \tau_{gen}$ we discussed previously.

In our conservative RF-EBM, the same finite-n contribution enters only through the Hadamard lift $A \odot (\cdot)$, and Corollary 4.3 shows that this lift removes the MP lower-edge mechanism by inflating the relevant ambient dimension from n to dn with $p/(dn) \to 0$. The resulting kernel spectrum has no vanishing band, so the linearized dynamics does not exhibit a separation between fast and slow directions driven by finite-n fluctuations. In other words, within this kernel-level model, there is no parametrically widening early-stopping window.

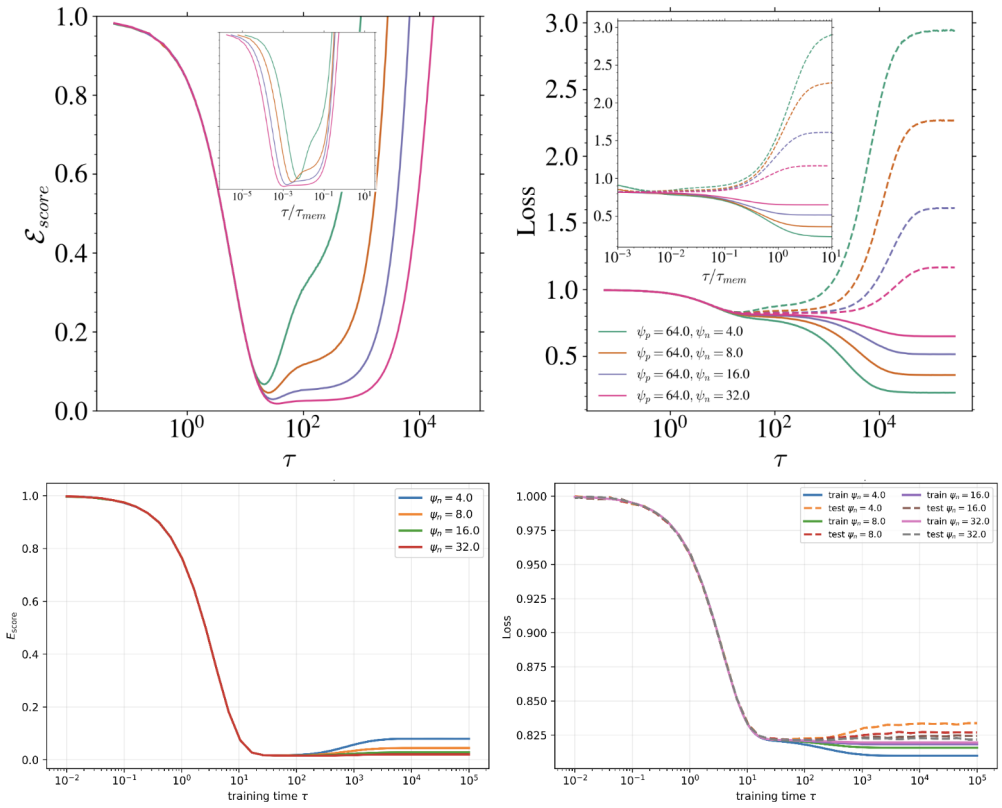

Figure 2: **Evolution of the training and test losses**. **Top** row shows results for random-feature diffusion model, from Bonnaire et al. **Bottom** row shows results for random-feature EBM (ours). For all four plots, the individual curves are for $\psi_n = 4, 8, 16, 32$, $\psi_p = 64$, $t = 0.1$ and $d = 100$. Both models exhibit the same initial generalization phase. However, diffusion score learning exhibits a pronounced memorization phase where E_{score} (distance to the true score) and test loss increase while training loss continues decreasing. This is consistent with an MP-driven small-eigenvalue band in U [2]. In constrast, EBM dynamics governed by H exhibits a negligible memorization phase: E_{score} and test loss do not increase even as training continues.

5 Experimental Results

Now, we validate our theory with empirical results.

Spectrum. In Figure 1, we reproduce the spectral regimes highlighted by [2] for U (two-scale spectrum in overparameterized random features), then compute the conservative kernel $H = A \odot U$ and verify the key qualitative change: the finite-n MP band in U does not translate into a comparable slow band in H in proportional scaling, and when $p \asymp dn$ we observe the predicted re-emergence of a nontrivial MP spectrum in the lifted dimension. Unless noted, we use $\sigma = \tanh$. This validates the mechanism behind Theorem 4.2.

Learning curves. In Figure 2, for the diffusion model, the broad spectrum of U yields a pronounced separation between a fast decrease for large eigenvalues (the generalization phase) and a slow tail for small eigenvalues (the memorization phase), reproducing the qualitative two-phase behavior of [2]. In contrast, the concentrated spectrum of H yields a much shorter tail and no n-growing slow phase driven by finite-n fluctuations, matching our theory in Section 4.4.

Deep EBM training Following [2], we train a deep EBM by score-matching and track training and test losses as a function of training timesteps. We use the CelebA dataset, which contains images of 32×32 generic faces.

We define $\tau^*(n)$ as the minimizer of test loss (the optimal early stopping time) and empirically study its scaling with n. Diffusion exhibits $\tau^*(n) \propto n$ in [2], as seen in Figure 3. However, we find instead that for DSM-trained EBMs $\tau^*(n)$ is sublinear and approximately constant, as shown in Figure 4. We observe the expected train/test divergence: training loss continues to decrease after test loss begins to rise, indicating memorization. Crucially, however, the *location* of the test-loss minimum does not move linearly with n in our EBM runs. Extracting $\tau^*(n)$ from the test-loss curves, we find that $\tau^*(n)$ grows sublinearly with n and is approximately constant over the tested range. In particular, plotting $\tau^*(n)$ against n shows far weaker scaling than the near-linear trend reported for diffusion models in [2]. Equivalently, plotting curves against τ/n does not produce the clean collapse characteristic of the diffusion mechanism.

The fact that there is a memorization phase differs from our random-feature case, where overfitting is absent. This is possibly due to the extreme overparametrization in training our deep EBM, since we used a large U-net architecture. We discuss this further in Section 6. On the other hand, the absence of an approximately linear $\tau^*(n)$ scaling in deep EBM training is consistent with the qualitative prediction that conservativity suppresses an n-dependent slow band and therefore prevents the emergence of an $O(n)$-widening generalization window as in the Bonnaire paper.

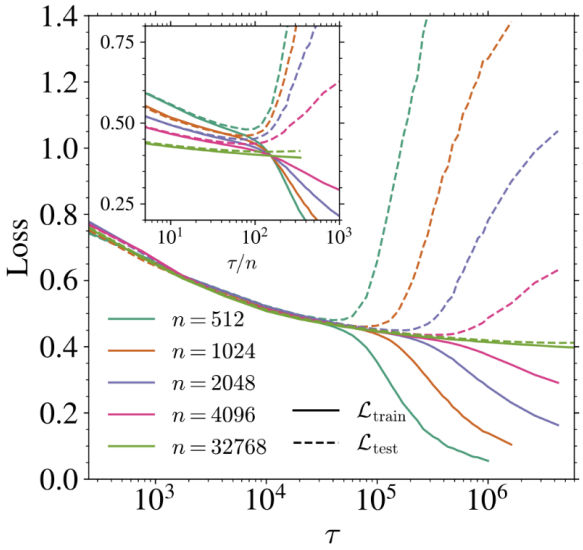

Figure 3: CelebA diffusion model: training loss (solid) and test loss (dashed) for varying training set sizes n. Overfitting occurs (train decreases while test increases), and the onset of this overscalign scales with n, as seen in the rescaled inset plot.

6 Conclusion

We studied generalization and memorization in score-trained generative models through the lens of generalization and memorization timescales. Building on the random-feature theory of [2], we showed EBMs do not have the $O(n)$ memorization timescale corresponding to the near-zero bulk of the feature covariance matrix U. Empirically, while deep EBMs trained by fixed-noise DSM still overfit on CelebA, the optimal early-stopping time $\tau^*(n)$ is constant, contrasting with the linear scaling seen for diffusion models. Together, these results suggest that diffusion models and EBMs are not reparameterizations of the same training dynamics: the conservative constraint for EBM scores changes the spectral structure that underlies implicit dynamical regularization, with direct consequences for training timescales.

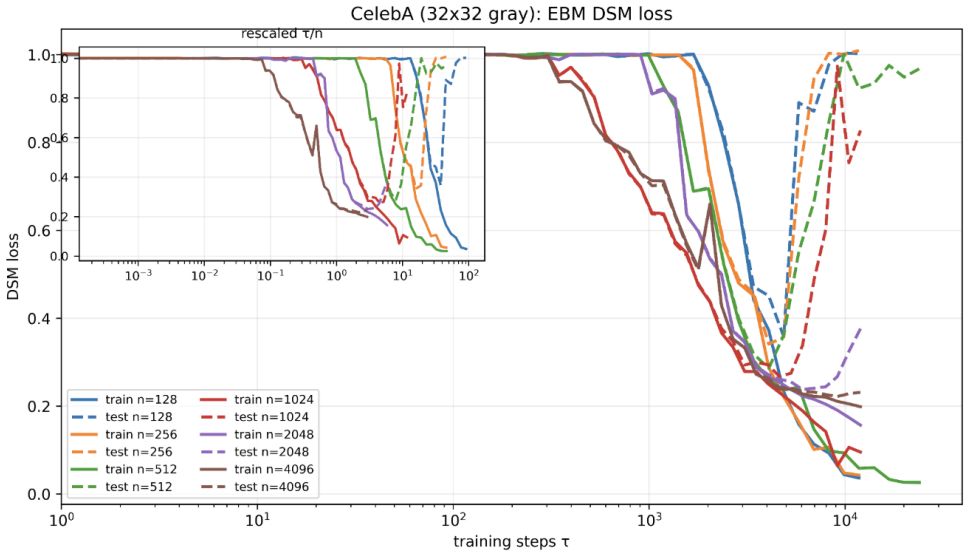

Figure 4: CelebA deep EBM: training loss (solid) and test loss (dashed) for varying training set sizes n. Overfitting occurs (train decreases while test increases), but the location of the test-loss minimum *does not* scale linearly with n.

6.1 Future work

Theorem 4.2 and Corollary 4.3 identify a specific mechanism by which the conservative dconstraint suppresses the finite-n MP band in the proportional regime: the Hadamard lift rewrites the finite-n term as a Gram matrix in ambient dimension dn, and when $p/(dn) \to 0$ this Gram matrix concentrates to I_p. This suggests a sharp structural transition outside the proportional regime. If p is scaled so that the lifted aspect ratio satisfies $p/(dn) \to \gamma \in (0, \infty)$ (in particular, if $n \asymp d$ this corresponds to $p \asymp d^2$), then the same Kronecker-lift identity shows the finite-n contribution becomes a genuine sample covariance matrix with nontrivial limiting spectrum. In that ultra-wide regime, one expects a Marchenko–Pastur band to reappear, with lower edge $(1 - \sqrt{\gamma})^2$; as $\gamma \uparrow 1$ this edge approaches 0 and produces an explicit family of slow modes in the linearized conservative DSM dynamics. *This would exactly explain the trend we see in Figure 4.* Establishing this rigorously (and characterizing how it interacts with the population term) would give a conservative analogue of the two-timescale picture of Bonnaire et al. in a width regime that is shifted from $p \asymp n$ to $p \asymp dn$ by the conservative constraint.

6.2 Declaration of Generative AI Use

Generative AI tools were used to assist with writing code and proofs. In particular, they provided help in fleshing out the algebra used and debugging the code.

References

[1] Mikhail Belkin et al. "Reconciling Modern Machine-Learning Practice and the Classical Bias–Variance Trade-Off". In: *Proceedings of the National Academy of Sciences* 116.32 (2019), pp. 15849–15854.

[2] Tony Bonnaire et al. "Why Diffusion Models Don't Memorize: The Role of Implicit Dynamical Regularization in Training". In: *arXiv preprint arXiv:2505.17638* (2025).

[3] Giovanni Catania et al. "A Theoretical Framework for Overfitting in Energy-Based Modeling". In: *arXiv preprint arXiv:2501.19158* (2025).

[4] Trevor Hastie et al. "Surprises in High-Dimensional Ridgeless Least Squares Interpolation". In: *The Annals of Statistics* 50.2 (2022), pp. 949–986.

[5] Jonathan Ho, Ajay Jain, and Pieter Abbeel. "Denoising Diffusion Probabilistic Models". In: *Advances in Neural Information Processing Systems*. arXiv:2006.11239. 2020.

[6] Aapo Hyvärinen. "Estimation of Non-Normalized Statistical Models by Score Matching". In: *Journal of Machine Learning Research* 6.24 (2005), pp. 695–709.

[7] Arthur Jacot, Franck Gabriel, and Clément Hongler. "Neural Tangent Kernel: Convergence and Generalization in Neural Networks". In: *Advances in Neural Information Processing Systems*. arXiv:1806.07572. 2018.

[8] Alethea Power et al. "Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets". In: *International Conference on Learning Representations*. arXiv:2201.02177. 2022.

[9] Jascha Sohl-Dickstein et al. "Deep Unsupervised Learning using Nonequilibrium Thermodynamics". In: *arXiv preprint arXiv:1503.03585* (2015).

[10] Yang Song and Diederik P. Kingma. "How to Train Your Energy-Based Models". In: *arXiv preprint arXiv:2101.03288* (2021).

[11] Yang Song et al. "Score-Based Generative Modeling through Stochastic Differential Equations". In: *International Conference on Learning Representations*. arXiv:2011.13456. 2021.

[12] Daniel Soudry et al. "The Implicit Bias of Gradient Descent on Separable Data". In: *Journal of Machine Learning Research* 19.70 (2018), pp. 1–57.

[13] Pascal Vincent. "A Connection Between Score Matching and Denoising Autoencoders". In: *Neural Computation* 23.7 (2011), pp. 1661–1674.

[14] Yuan Yao, Lorenzo Rosasco, and Andrea Caponnetto. "On Early Stopping in Gradient Descent Learning". In: *Constructive Approximation* (2007). Preprint widely circulated as `earlystop.pdf`.

A Proofs for Sections 3 and 4

This appendix contains proofs of the auxiliary concentration tools used in the main text and of the main claims in Section 4. We do not prove the Gaussian-equivalence surrogate (Lemma 3.2) since it is proved in detail in Bonnaire et al. Before we give these proofs, we first note some standard results that we use several times in the course of our proofs and that we take for granted.

1. **Rayleigh quotient / operator norm.** For symmetric B, $\|B\|_{\mathrm{op}} = \sup_{\|u\|_2=1} |u^\top Bu|$, and $\lambda_{\min}(B) = \min_{\|u\|_2=1} u^\top Bu$, $\lambda_{\max}(B) = \max_{\|u\|_2=1} u^\top Bu$.
2. **Weyl's inequality.** For symmetric M, N and each k, $|\lambda_k(M) - \lambda_k(N)| \le \|M - N\|_{\mathrm{op}}$.
3. **E-nets on spheres.** For $\varepsilon \in (0,1)$, there exists ε-net $\mathcal{M}_\varepsilon \subset S^{m-1}$ with $|\mathcal{M}_\varepsilon| \le (1+2/\varepsilon)^m$, and for any $x \in \mathbb{R}^m$,
$$\|x\|_2 \le \frac{1}{1-\varepsilon} \max_{v \in \mathcal{M}_\varepsilon} \langle v, x\rangle.$$
4. **Bernstein inequality for sub-exponential variables.** If $X_1, \dots, X_m$ are independent, mean zero, with $\|X_i\|_{\psi_1} \le K$, then there exist $c, C > 0$ depending only on K such that for all $t \ge 0$,
$$\mathbb{P}\Big(\Big|\sum_{i=1}^m X_i\Big| \ge t\Big) \le 2\exp\Big(-c\min\{t^2/(CK^2m),\ t/(CK)\}\Big).$$

A.1 Proof of Lemma 3.5

Proof of Lemma 3.5. Let $Z \in \mathbb{R}^{p\times D}$ have independent rows $z_1^\top, \dots, z_p^\top$ satisfying the assumptions of Lemma 3.5. Set
$$G := \frac{1}{D} ZZ^\top \in \mathbb{R}^{p\times p}.$$
Since G is symmetric, its operator norm is given by the Rayleigh quotient,
$$\|G - I_p\|_{\mathrm{op}} = \sup_{\|u\|_2=1} \big|u^\top (G - I_p) u\big|. \tag{22}$$
Fix $u \in \mathbb{R}^p$ with $\|u\|_2 = 1$ and write
$$S(u) := Z^\top u = \sum_{i=1}^p u_i z_i \in \mathbb{R}^D.$$
Then
$$u^\top G u = \frac{1}{D} u^\top ZZ^\top u = \frac{1}{D}\|Z^\top u\|_2^2 = \frac{1}{D}\|S(u)\|_2^2. \tag{23}$$
By isotropy and independence of the rows,
$$\mathbb{E}[S(u)] = \sum_{i=1}^p u_i\, \mathbb{E}[z_i] = 0, \qquad \mathbb{E}[S(u)S(u)^\top] = \sum_{i=1}^p u_i^2\, \mathbb{E}[z_i z_i^\top] = \sum_{i=1}^p u_i^2 I_D = I_D,$$
and therefore
$$\mathbb{E}\|S(u)\|_2^2 = \mathrm{Tr}(\mathbb{E}[S(u)S(u)^\top]) = \mathrm{Tr}(I_D) = D. \tag{24}$$
Combining (23)–(24), we have $\mathbb{E}[u^\top G u] = 1$ for every unit u.

We now record the one-dimensional sub-exponential control inherited by $S(u)$. For any unit $v \in \mathbb{R}^D$,
$$\langle v, S(u)\rangle = \sum_{i=1}^p u_i \langle v, z_i\rangle.$$
The r.v.s $\langle v, z_i\rangle$ are independent across i, centered, and satisfy $\|\langle v, z_i\rangle\|_{\psi_1} \le \|z_i\|_{\psi_1} \le K$, so by the Bernstein inequality, there exist $c_0, C_0 > 0$ (depending only on K) such that for all $s \ge 0$,
$$\mathbb{P}\Big(\big|\langle v, S(u)\rangle\big| \ge s\Big) \le 2\exp\Big(-c_0\ \min\{s^2, s\}\Big). \tag{25}$$

From (25) one obtains concentration for the Euclidean norm of $S(u)$ at scale $\sqrt{D}$ as follows. Let $\mathcal{M}_\varepsilon \subset S^{D-1}$ be an ε-net of the unit sphere, with cardinality bounded by $|\mathcal{M}_\varepsilon| \le (1+2/\varepsilon)^D$. For any $x \in \mathbb{R}^D$ and any $\varepsilon \in (0,1)$,

$$\|x\|_2 \le \frac{1}{1-\varepsilon} \max_{v \in \mathcal{M}_\varepsilon} \langle v, x\rangle, \qquad \|x\|_2 \ge \max_{v \in \mathcal{M}_\varepsilon} \langle v, x\rangle. \tag{26}$$

(Indeed, for $\|x\|_2 \neq 0$, pick $v \in \mathcal{M}_\varepsilon$ with $\|v - x/\|x\|_2\|_2 \le \varepsilon$ and use $\langle v, x\rangle = \|x\|_2 \langle v, x/\|x\|_2\rangle \ge (1-\varepsilon)\|x\|_2$; the upper bound follows similarly.)

Fix $\varepsilon = 1/4$ so that $|\mathcal{N}_{1/4}| \le 9^D$ and $(1-\varepsilon)^{-1} = 4/3$. Using (26) with $x = S(u)$ and a union bound over $v \in \mathcal{N}_{1/4}$, we obtain for all $r \ge 0$,

$$\begin{aligned}
\mathbb{P}\big(\|S(u)\|_2 \ge \sqrt{D} + r\big) &\le \mathbb{P}\Big(\max_{v \in \mathcal{N}_{1/4}} \langle v, S(u)\rangle \ge \tfrac{3}{4}(\sqrt{D} + r)\Big) \\
&\le \sum_{v \in \mathcal{N}_{1/4}} \mathbb{P}\Big(\langle v, S(u)\rangle \ge \tfrac{3}{4}(\sqrt{D} + r)\Big) \\
&\le 9^D \cdot 2\exp\Big(-c_0 \min\big\{(\sqrt{D}+r)^2,\ \sqrt{D}+r\big\}\Big).
\end{aligned} \tag{27}$$

Similarly, by (26) and the union bound,

$$\begin{aligned}
\mathbb{P}\big(\|S(u)\|_2 \le \sqrt{D} - r\big) &\le \mathbb{P}\Big(\max_{v \in \mathcal{N}_{1/4}} \langle v, S(u)\rangle \le \sqrt{D} - r\Big) \\
&\le \sum_{v \in \mathcal{N}_{1/4}} \mathbb{P}\Big(\langle v, S(u)\rangle \le -(\sqrt{D} - r)\Big) \\
&\le 9^D \cdot 2\exp\Big(-c_0 \min\big\{(\sqrt{D}-r)^2,\ \sqrt{D}-r\big\}\Big).
\end{aligned} \tag{28}$$

Combining (27)–(28) and using that, for $0 \le r \le \sqrt{D}/2$,

$$(\sqrt{D} \pm r)^2 = D \pm 2r\sqrt{D} + r^2 \ge D + r^2, \qquad \sqrt{D} \pm r \ge \frac{\sqrt{D}}{2},$$

we obtain constants $c_1, C_1 > 0$ (depending only on K) such that for all $0 \le r \le \sqrt{D}/2$,

$$\mathbb{P}\Big(\big|\|S(u)\|_2 - \sqrt{D}\big| \ge r\Big) \le 2\exp\Big(-c_1 r^2\Big). \tag{29}$$

(For $r > \sqrt{D}/2$ the conclusion (9) is trivial after enlarging constants, since the right-hand side becomes larger than 1 while $\|G - I_p\|_{\mathrm{op}} \le \|G\|_{\mathrm{op}} + 1 \le \frac{1}{D}\|Z\|_{\mathrm{op}}^2 + 1$ is finite almost surely.)

We now translate (29) into concentration for the quadratic form $u^\top (G - I_p) u$. Since

$$\frac{1}{D}\|S(u)\|_2^2 - 1 = \Big(\frac{\|S(u)\|_2}{\sqrt{D}} - 1\Big)\Big(\frac{\|S(u)\|_2}{\sqrt{D}} + 1\Big),$$

the implication

$$\Big|\frac{1}{D}\|S(u)\|_2^2 - 1\Big| \ge \alpha \quad \Longrightarrow \quad \Big|\|S(u)\|_2 - \sqrt{D}\Big| \ge \frac{\alpha}{3}\sqrt{D}$$

holds whenever $\alpha \in (0,1)$ and $\|S(u)\|_2 \le 2\sqrt{D}$ (since then $\|S(u)\|_2/\sqrt{D} + 1 \le 3$). Applying (29) with $r = \frac{\alpha}{3}\sqrt{D}$ yields, for $\alpha \in (0,1)$,

$$\mathbb{P}\Big(\Big|\frac{1}{D}\|S(u)\|_2^2 - 1\Big| \ge \alpha\Big) \le 2\exp\Big(-c_2 D\alpha^2\Big) \tag{30}$$

for some $c_2 > 0$ depending only on K.

We now make the bound uniform over $u \in S^{p-1}$ by a net argument in $\mathbb{R}^p$. Let $_{\mathrm{d}} \subset S^{p-1}$ be a d-net with $|_{\mathrm{d}}| \le (1 + 2/\mathrm{d})^p$. For any symmetric matrix $B \in \mathbb{R}^{p \times p}$ and any $\mathrm{d} \in (0, 1/2)$,

$$\|B\|_{\mathrm{op}} \le \frac{1}{1 - 2\mathrm{d}} \max_{u \in_{\mathrm{d}}} |u^\top B u|. \tag{31}$$

To verify (31), fix $\|x\|_2 = 1$ and choose $u \in_{\mathrm{d}}$ with $\|x - u\|_2 \le \mathrm{d}$. Then

$$|x^\top Bx - u^\top Bu| = |(x-u)^\top Bx + u^\top B(x-u)| \le \|x-u\|_2 \, \|B\|_{\mathrm{op}} \, \|x\|_2 + \|u\|_2 \, \|B\|_{\mathrm{op}} \, \|x-u\|_2 \le 2\mathrm{d} \, \|B\|_{\mathrm{op}},$$

and therefore $|x^\top Bx| \le |u^\top Bu| + 2\mathrm{d}\|B\|_{\mathrm{op}}$. Taking the supremum over $\|x\|_2 = 1$ gives $\|B\|_{\mathrm{op}} \le \max_{u \in_{\mathrm{d}}} |u^\top Bu| + 2\mathrm{d}\|B\|_{\mathrm{op}}$, which rearranges to (31).

We apply (31) to $B = G - I_p$ with $\mathrm{d} = 1/4$, so $(1 - 2\mathrm{d})^{-1} = 2$:

$$\|G - I_p\|_{\mathrm{op}} \le 2 \max_{u \in_{1/4}} \Big| \frac{1}{D} \|S(u)\|_2^2 - 1 \Big|.$$

Fix $t \ge 0$ and set

$$\alpha := C\Big(\sqrt{\frac{p}{D}} + \frac{t}{\sqrt{D}}\Big)$$

with $C > 0$ chosen later (depending only on K). For D large enough, $\alpha \in (0, 1)$ since $p/D \to 0$ in the regime of interest; in any case, if $\alpha \ge 1$ the claimed inequality is trivial after enlarging constants. By the union bound, (30), and $|_{1/4}| \le 9^p$,

$$\begin{aligned}
\mathbb{P}\big(\|G - I_p\|_{\mathrm{op}} \ge 2\alpha\big) &\le \mathbb{P}\Big(\max_{u \in_{1/4}} \Big| \frac{1}{D} \|S(u)\|_2^2 - 1 \Big| \ge \alpha \Big) \\
&\le \sum_{u \in_{1/4}} \mathbb{P}\Big(\Big| \frac{1}{D} \|S(u)\|_2^2 - 1 \Big| \ge \alpha \Big) \\
&\le 9^p \cdot 2 \exp\big(- c_2 D \alpha^2 \big).
\end{aligned}$$

Since $D\alpha^2 = C^2(p + t^2)$, we may choose C large enough so that $c_2 C^2 \ge 2 \log 9$; then

$$9^p \cdot 2 \exp\big(- c_2 D \alpha^2 \big) \le 2 \exp(-ct^2)$$

for some $c > 0$ depending only on K. Hence, with probability at least $1 - 2e^{-ct^2}$,

$$\|G - I_p\|_{\mathrm{op}} \le 2\alpha = C\Big(\sqrt{\frac{p}{D}} + \frac{t}{\sqrt{D}}\Big), \tag{32}$$

after adjusting C.

Finally, we convert (32) into the exact form (9). Let $s_{\max}(Z)$ and $s_{\min}(Z)$ denote the largest and smallest singular values of Z. The eigenvalues of $G = (1/D)ZZ^\top$ are $\{s_k(Z)^2/D\}_{k=1}^p$, so

$$\|G - I_p\|_{\mathrm{op}} = \max\Big\{ \frac{s_{\max}(Z)^2}{D} - 1, \; 1 - \frac{s_{\min}(Z)^2}{D} \Big\}.$$

Define

$$\delta := C\Big(\sqrt{\frac{p}{D}} + \frac{t}{\sqrt{D}}\Big),$$

with the same C as above. The bound (32) implies (after enlarging C by an absolute factor if needed) that, with probability at least $1 - 2e^{-ct^2}$,

$$1 - \delta \le \frac{s_{\min}(Z)}{\sqrt{D}} \le \frac{s_{\max}(Z)}{\sqrt{D}} \le 1 + \delta.$$

Therefore, on this event,

$$\begin{aligned}
\|G - I_p\|_{\mathrm{op}} &\le \max\{(1+\delta)^2 - 1, \; 1 - (1-\delta)^2\} \\
&= \max\{2\delta + \delta^2, \; 2\delta - \delta^2\} \le 2\delta + \delta^2 \\
&\le C\Big(\sqrt{\frac{p}{D}} + \frac{t}{\sqrt{D}}\Big) + C^2\Big(\frac{p}{D} + \frac{t^2}{D}\Big),
\end{aligned}$$

after adjusting constants once more. This is exactly (9). The final claim follows by taking t fixed and letting $p/D \to 0$, which forces the right-hand side to vanish in probability. □

A.2 Proof of Proposition 4.1

Proof. Recall that for each ν we write $\phi^\nu := \sigma(Wy^\nu/\sqrt{d}) \in \mathbb{R}^p$, so that

$$\psi_\theta(y^\nu) = \frac{1}{\sqrt{p}} W^\top(\theta \odot \phi^\nu) = \frac{1}{\sqrt{p}} W^\top \mathrm{diag}(\phi^\nu)\theta.$$

Expand the square in (12):

$$L_{\text{train}}(\theta) = \frac{1}{dn}\sum_{\nu=1}^{n} \mathbb{E}_{\xi^\nu}\|\psi_\theta(y^\nu)\|_2^2 + \frac{2}{dn}\sum_{\nu=1}^{n} \mathbb{E}_{\xi^\nu}\Big\langle \psi_\theta(y^\nu), \frac{1}{\sqrt{\tau}}\xi^\nu \Big\rangle + \frac{1}{dn}\sum_{\nu=1}^{n} \mathbb{E}_{\xi^\nu}\Big\| \frac{1}{\sqrt{\tau}}\xi^\nu \Big\|_2^2.$$

For the quadratic term, using $\psi_\theta(y^\nu) = \frac{1}{\sqrt{p}} W^\top \mathrm{diag}(\phi^\nu)\theta$,

$$\|\psi_\theta(y^\nu)\|_2^2 = \frac{1}{p}\theta^\top \mathrm{diag}(\phi^\nu)\, WW^\top\, \mathrm{diag}(\phi^\nu)\theta.$$

Let $A := WW^\top/d$. Entrywise,

$$\big(\mathrm{diag}(\phi^\nu)\, A\, \mathrm{diag}(\phi^\nu)\big)_{ij} = \phi_i^\nu\, A_{ij}\, \phi_j^\nu = \big(A \odot (\phi^\nu(\phi^\nu)^\top)\big)_{ij},$$

so $\mathrm{diag}(\phi^\nu)\, A\, \mathrm{diag}(\phi^\nu) = A \odot (\phi^\nu(\phi^\nu)^\top)$. Hence

$$\|\psi_\theta(y^\nu)\|_2^2 = \frac{d}{p}\theta^\top \big(A \odot (\phi^\nu(\phi^\nu)^\top)\big)\theta,$$

and therefore

$$\begin{aligned}
\frac{1}{dn}\sum_{\nu=1}^{n} \mathbb{E}\|\psi_\theta(y^\nu)\|_2^2 &= \frac{1}{dn}\sum_{\nu=1}^{n} \mathbb{E}\Big[\frac{d}{p}\theta^\top \big(A \odot (\phi^\nu(\phi^\nu)^\top)\big)\theta\Big] \\
&= \frac{1}{p}\theta^\top \Big(A \odot \frac{1}{n}\sum_{\nu=1}^{n} \mathbb{E}[\phi^\nu(\phi^\nu)^\top]\Big)\theta \\
&= \frac{1}{p}\theta^\top H_{\text{train}}\theta.
\end{aligned}$$

For the linear term,

$$\begin{aligned}
\Big\langle \psi_\theta(y^\nu), \frac{1}{\sqrt{\tau}}\xi^\nu \Big\rangle &= \frac{1}{\sqrt{\tau}}\Big\langle \frac{1}{\sqrt{p}} W^\top \mathrm{diag}(\phi^\nu)\theta,\ \xi^\nu \Big\rangle = \frac{1}{\sqrt{p\tau}}\Big\langle \mathrm{diag}(\phi^\nu)\theta,\ W\xi^\nu \Big\rangle \\
&= \frac{1}{\sqrt{p\tau}}\sum_{i=1}^{p} \theta_i \phi_i^\nu (w_i^\top \xi^\nu).
\end{aligned}$$

Thus

$$\begin{aligned}
\frac{2}{dn}\sum_{\nu=1}^{n} \mathbb{E}\Big\langle \psi_\theta(y^\nu), \frac{1}{\sqrt{\tau}}\xi^\nu \Big\rangle &= \frac{2}{dn}\sum_{\nu=1}^{n} \frac{1}{\sqrt{p\tau}}\sum_{i=1}^{p} \theta_i\, \mathbb{E}[\phi_i^\nu (w_i^\top \xi^\nu)] \\
&= \frac{2}{\sqrt{p}}\sum_{i=1}^{p} \theta_i \Big(\frac{1}{nd\sqrt{\tau}}\sum_{\nu=1}^{n} \mathbb{E}[\phi_i^\nu (w_i^\top \xi^\nu)]\Big) = \frac{2}{\sqrt{p}}\theta^\top b_{\text{train}}.
\end{aligned}$$

For the constant term, $\mathbb{E}\|\xi^\nu\|_2^2 = d$, so

$$\frac{1}{dn}\sum_{\nu=1}^{n} \mathbb{E}\Big\| \frac{1}{\sqrt{\tau}}\xi^\nu \Big\|_2^2 = \frac{1}{dn}\cdot n \cdot \frac{1}{\tau}\cdot d = \frac{1}{\tau}.$$

Combining the quadratic, linear, and constant terms proves (15). □

A.3 Proof of Theorem 4.2

Proof. By hypothesis, using Lemma 3.2, we may write

$$U_{\text{train}} = s^2 I_p + v^2 \frac{1}{n}\Omega\Omega^\top + b^2 A + E, \qquad \|E\|_{\text{op}} \xrightarrow{\mathbb{P}} 0, \qquad A = \frac{1}{d}WW^\top.$$

Then

$$H_{\text{train}} = A \odot U_{\text{train}} = s^2(A \odot I_p) + v^2\Big(A \odot \frac{1}{n}\Omega\Omega^\top\Big) + b^2(A \odot A) + (A \odot E).$$

We control each term in the operator norm.

Diagonal term. Since $A_{ii} = \|w_i\|_2^2/d$ with $\|w_i\|_2^2 \sim \chi_d^2$, for any fixed $\varepsilon \in (0,1)$,

$$\mathbb{P}(|A_{ii} - 1| \geq \varepsilon) \leq 2\exp(-cd\varepsilon^2)$$

for a universal $c > 0$ by χ^2 concentration. Since $p \asymp d$, a union bound gives

$$\max_{i \leq p} |A_{ii} - 1| \xrightarrow{\mathbb{P}} 0,$$

hence $\|\text{diag}(A) - I_p\|_{\text{op}} \xrightarrow{\mathbb{P}} 0$. Since $A \odot I_p = \text{diag}(A)$, this means $s^2(A \odot I_p) = s^2 I_p + o_{\mathbb{P}}(1)$.

Schur-lifted finite-n term. By Lemma 3.4,

$$A \odot \frac{1}{n}\Omega\Omega^\top = \frac{1}{dn}ZZ^\top, \qquad z_i = w_i \otimes \omega_i \in \mathbb{R}^{dn},$$

where $\omega_i^\top$ is the ith row of Ω. The pairs (w_i, ω_i) are i.i.d. across i, hence so are the rows z_i. Also $\mathbb{E}[z_i] = 0$, and for indices $(a,b), (a',b') \in [d] \times [n]$,

$$\mathbb{E}[z_{i,(a,b)} z_{i,(a',b')}] = \mathbb{E}[w_{i,a} w_{i,a'}]\mathbb{E}[\omega_{i,b}\omega_{i,b'}] = \delta_{aa'}\delta_{bb'},$$

so $\mathbb{E}[z_i z_i^\top] = I_{dn}$. Finally, each coordinate $w_{i,a}\omega_{i,b}$ is a product of independent standard Gaussians and is sub-exponential, so $\|z_i\|_{\psi_1} \leq K$ for a universal K. Lemma 3.5 therefore applies with $D = dn$. Since $p/(dn) \to 0$ in the proportional regime, we obtain

$$\Big\|A \odot \frac{1}{n}\Omega\Omega^\top - I_p\Big\|_{\text{op}} \xrightarrow{\mathbb{P}} 0.$$

Hadamard square. Define $q_i := w_i \otimes w_i \in \mathbb{R}^{d^2}$ and let Q have rows $q_i^\top$. Then

$$(A^{\odot 2})_{ij} = \Big(\frac{w_i^\top w_j}{d}\Big)^2 = \frac{1}{d^2} q_i^\top q_j, \qquad \text{so} \qquad A^{\odot 2} = \frac{1}{d^2}QQ^\top.$$

Let $\mu := \mathbb{E}[q_i]$. Since $\mathbb{E}[w_{i,a}w_{i,a'}] = \delta_{aa'}$, we have $\mu = \text{vec}(I_d)$ and $\|\mu\|_2^2 = \|I_d\|_F^2 = d$. Write $q_i = \mu + \tilde{q}_i$ with $\mathbb{E}[\tilde{q}_i] = 0$ and let $\tilde{Q}$ have rows $\tilde{q}_i^\top$. Then

$$\frac{1}{d^2}QQ^\top = \frac{1}{d^2}\tilde{Q}\tilde{Q}^\top + \frac{\|\mu\|_2^2}{d^2}\mathbf{1}\mathbf{1}^\top + \frac{1}{d^2}\Big(\tilde{Q}\mu\,\mathbf{1}^\top + \mathbf{1}(\tilde{Q}\mu)^\top\Big).$$

For the centered Gram term, compute

$$\mathbb{E}\|\tilde{q}_i\|_2^2 = \sum_{a,a'} \mathbb{E}[(w_a w_{a'} - \delta_{aa'})^2] = \sum_{a=1}^{d} \mathbb{E}[(w_a^2 - 1)^2] + \sum_{a \neq a'} \mathbb{E}[(w_a w_{a'})^2] = 2d + (d^2 - d) = d^2 + d,$$

so $\mathbb{E}[(1/d^2)\tilde{Q}\tilde{Q}^\top] = (1 + 1/d)I_p$. Moreover $\tilde{q}_i$ has sub-exponential coordinates (centered quadratic monomials of Gaussians). Since $p/d^2 \to 0$ when $p \asymp d$, Lemma 3.5 applied to the normalized rows $\tilde{q}_i/\sqrt{1 + 1/d}$ in ambient dimension $D = d^2$ yields

$$\Big\|\frac{1}{d^2}\tilde{Q}\tilde{Q}^\top - \Big(1 + \frac{1}{d}\Big)I_p\Big\|_{\text{op}} \xrightarrow{\mathbb{P}} 0.$$

For the middle term, since $\|\mu\|_2^2 = \|I_d\|_F^2 = d$, we easily get that $(\|\mu\|_2^2/d^2)\mathbf{1}\mathbf{1}^\top = (1/d)\mathbf{1}\mathbf{1}^\top$. Finally, the cross terms are negligible. Indeed, $(\tilde{Q}\mu)_i = \tilde{q}_i^\top \mu = \sum_{a=1}^{d}(w_{i,a}^2 - 1)$ has variance $2d$, so $\|\tilde{Q}\mu\|_2 = O_{\mathbb{P}}(\sqrt{pd})$ and therefore

$$\Big\|\frac{1}{d^2}\tilde{Q}\mu\,\mathbf{1}^\top\Big\|_{\text{op}} \leq \frac{1}{d^2}\|\tilde{Q}\mu\|_2\,\|\mathbf{1}\|_2 = \frac{1}{d^2}\,O_{\mathbb{P}}(\sqrt{pd}) \cdot \sqrt{p} = O_{\mathbb{P}}\Big(\frac{p}{d^{3/2}}\Big) = o_{\mathbb{P}}(1),$$

since $p \asymp d$. The same holds for its transpose. Hence

$$A^{\odot 2} = \Big(1 + \frac{1}{d}\Big)I_p + \frac{1}{d}\mathbf{1}\mathbf{1}^\top + o_{\mathbb{P}}(1).$$

Surrogate error. Since $A \succeq 0$, the Schur product theorem implies $A \odot |E| \succeq 0$ and the standard bound $\|A \odot E\|_{\mathrm{op}} \leq \|A\|_{\mathrm{op}} \|E\|_{\mathrm{op}}$ holds. Moreover $\|A\|_{\mathrm{op}} = O_{\mathbb{P}}(1)$ when $p/d \to \psi_p \in (0, \infty)$ (e.g. by the Marchenko–Pastur bound for $WW^\top/d$). Hence $\|A \odot E\|_{\mathrm{op}} = o_{\mathbb{P}}(1)$.

Combine. Collecting the bounds,

$$H_{\mathrm{train}} = s^2 I_p + v^2 I_p + b^2 I_p + \frac{b^2}{d} \mathbf{1}\mathbf{1}^\top + o_{\mathbb{P}}(1) = (s^2 + v^2 + b^2) I_p + \frac{b^2}{d} \mathbf{1}\mathbf{1}^\top + o_{\mathbb{P}}(1),$$

in operator norm, which is exactly (19). □

A.4 Proof of Corollary 4.3

Proof. Let $c := s^2 + v^2 + b^2$ and $R := \frac{b^2}{d} \mathbf{1}\mathbf{1}^\top$. By Theorem 4.2,

$$\|H_{\mathrm{train}} - (cI_p + R)\|_{\mathrm{op}} = o_{\mathbb{P}}(1).$$

By Weyl's inequality, for each k,

$$\left|\lambda_k(H_{\mathrm{train}}) - \lambda_k(cI_p + R)\right| \leq \|H_{\mathrm{train}} - (cI_p + R)\|_{\mathrm{op}},$$

and therefore $\max_{k \leq p} |\lambda_k(H_{\mathrm{train}}) - \lambda_k(cI_p + R)| = o_{\mathbb{P}}(1)$.

Since R has rank one, $\mathbf{1}$ is an eigenvector with eigenvalue $\lambda_{\max}(R) = \frac{b^2}{d} \|\mathbf{1}\|_2^2 = \frac{b^2}{d} p = b^2 \psi_p + o(1)$, and R vanishes on $\mathbf{1}^\perp$. Hence the spectrum of $cI_p + R$ consists of one eigenvalue $c + \lambda_{\max}(R)$ and $p - 1$ eigenvalues equal to c. This yields the first two displayed claims.

Assume $c > 0$. Since $\|H_{\mathrm{train}} - (cI_p + R)\|_{\mathrm{op}} = o_{\mathbb{P}}(1)$, with probability tending to 1 we have $\|H_{\mathrm{train}} - (cI_p + R)\|_{\mathrm{op}} \leq c/2$. On this event, for any unit vector $u \in \mathbb{R}^p$,

$$\begin{aligned} u^\top H_{\mathrm{train}} u &= u^\top (cI_p + R) u + u^\top (H_{\mathrm{train}} - (cI_p + R)) u \\ &\geq u^\top (cI_p + R) u - \|H_{\mathrm{train}} - (cI_p + R)\|_{\mathrm{op}} \\ &\geq c - \frac{c}{2} \\ &= \frac{c}{2}, \end{aligned}$$

since $R \succeq 0$ implies $u^\top (cI_p + R) u \geq c$. Taking the minimum over $\|u\|_2 = 1$ and using the Rayleigh quotient gives that on the event

$$\lambda_{\min}(H_{\mathrm{train}}) = \min_{\|u\|_2 = 1} u^\top H_{\mathrm{train}} u \geq c/2 > 0,$$

which is what we wanted to show. □